

ASYMPTOTIC MATCHING AND INITIAL-SET DEPENDENCE IN THE PRECISION COST OF CONTRACTING CONTROL SYSTEMS

AUTHOR INFORMATION PENDING

ABSTRACT. We study the minimum number of open-loop controls required to keep every initial state within a Euclidean tolerance of a constraint set at all times of a finite horizon. For an explicit class of smooth planar contracting systems, feasible branch lengths and radial contraction may depend independently on the state. Matching at the common attracting origin, together with directly checkable small derivative bounds, yields a common-cost boundary $\gamma_* = \beta h_a$, where $h_a = -\sum_i a_i \log a_i$ and $\delta_n = \exp(-\gamma n + o(n))$. At and below this boundary, every fixed positive-area compact initial set has rate γ/β . Above it, one compact set of arbitrarily nearly full area has rate h_a for all finite higher exponents and their precision sequences, whereas the full constraint set has a strictly larger lower rate. The proof compares arbitrary approximate controls with stopped physical witnesses at a loss of at most $1 + Bn$. It does not require an instantaneous power identity between branch lengths and radial factors. We also give instances excluding coordinate reduction to specified exactly matched models. A separate radius-autonomous mismatch theorem and a smooth collapse example delimit the matching condition and the role of global controlled geometry.

1. INTRODUCTION

Invariance entropy measures the control information required to keep an initial set inside a controlled invariant set; see Kawan [8]. When all controls contract toward a point, outer invariance complexity can be bounded at every fixed positive tolerance. An exponentially decreasing tolerance can nevertheless require exponentially many controls. We ask when this precision cost is shared by all fixed positive-area compact initial sets, and when the geometry of the initial set changes its rate.

The catalogue used here assigns an open-loop input to each actual initial state. One tolerance δ_n applies at every integer time from 0 through n . A trajectory may leave and return to the constraint set within that tolerance. For this task, neither an exact coding nor convergence of all trajectories alone determines the optimal catalogue. An almost full-area set may exclude the rare states which determine the full-set cost.

Our main result treats two complementary feasible branches with limiting proportions a_0, a_1 and radial first derivatives a_0^β, a_1^β . The branch proportions and radial factors are independent smooth functions of the state. A smallness condition on these functions gives complete feasible intervals, even though states with the same initial radius and control word may have different subsequent radial trajectories. Their cumulative angular and radial errors are bounded separately. Thus instantaneous matching is unnecessary: the boundary βh_a survives this specified class of perturbations.

The controlled geometric step concerns arbitrary approximate inputs. A logarithmic slope cone governs the moving feasible graphs. At a first wrong branch, the physical excess outside the constraint yields a one-sided band in the actual initial fibre. Complete stopped intervals provide separated witnesses, and one input covers at most $1 + Bn$ of them. A

2020 *Mathematics Subject Classification.* Primary 93C55, 37B40; Secondary 93D15, 37C05.

Key words and phrases. Outer invariance complexity, precision cost, controlled contraction, asymptotic matching, feasible geometry, initial-set dependence.

global bound toward the origin supplies the entire tail after an early stop. Only after these estimates have been proved do classical stopping and type counts give the precision rates.

Theorem 2.2 proves the common-cost boundary. A separate radius-autonomous family, Theorem 2.3, shows that unequal radial matching powers cause initial-set dependence at every sufficiently small positive exponent. This necessity statement stays within that family. Section 7 gives instances of the main theorem which cannot be reduced to the specified previous forms by a simultaneous change of coordinates preserving the constraint. Section 8 shows that agreement near the origin and all-input convergence, without the global geometric conditions, do not imply the boundary.

1.1. Relation to previous work. The finite-horizon count below is the outer invariance complexity of Chen and Zhong [1, Section 2.2], with $n + 1$ constrained times instead of n . Their fixed-tolerance complexity and equi-invariability results provide the underlying catalogue framework. Exponential and practical stabilization entropies [3, 4] address related information costs for different stabilization tasks. We do not introduce a new catalogue definition.

Measure-theoretic invariance entropies and variational principles are developed in [10, 9]. The entropy dimensions of Chen, Huang and Zhong [2] use invariant-partition cylinders and weights $\exp[-s(N\tau)^\alpha]$, with a separate measure-covering tolerance. Their cylinder and variational estimates do not by themselves control the Euclidean first-deviation bands of arbitrary inputs at δ_n . On our connected Q , a finite clopen invariant partition would have only one nonempty cell, while neither control preserves all of Q . The main clopen variational theorems therefore do not directly apply. Measurable-cylinder estimates can still be useful, but require the physical comparison proved here. Ordinary area need not be invariant; its use is not a new measure-theoretic framework.

Graph-transform and volume-distortion methods are existing tools. Kawan and Da Silva [7, Appendix] prove volume estimates near a partially hyperbolic all-time lift. For our contracting systems, every bounded bi-infinite trajectory is the origin: $\|x_0\|_* \leq L^k \sup_j \|x_j\|_*$ for all k . The physical angular coordinate is singular there. Those local volume estimates alone do not give the complete feasible splits or the arbitrary-input comparison as the tolerance decreases with the horizon.

Prefix stopping, product weights, types, and typical sets are classical [6, 5]. Yang, Chen, Yang and Zhou [11] develop cost-threshold pressure and Bowen-equation results on a given invariant partition. Their spanning/separated comparison preserves the prescribed address; it does not compare that address with every approximately feasible physical input. The additional step needed here is the band estimate and stopped-witness comparison of Section 4. These comparisons with [7, 11] use the graph-transform appendix and the author versions arXiv:1711.01181v2 and arXiv:2309.01628v3, respectively. Our proofs of the required estimates are self-contained.

2. COUNTING PROBLEM AND MAIN RESULTS

Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For two maps $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $u = (u_1, u_2, \dots) \in \{0, 1\}^\mathbb{N}$, write $F_u^0 x = x$ and $F_u^k x = F_{u_k} \circ \dots \circ F_{u_1} x$. All binary inputs are allowed. Distances are Euclidean, and $Q^\delta = \{x : \text{dist}(x, Q) < \delta\}$.

Definition 2.1. For a nonempty compact $K \subset Q$, let $r_F(n, \delta; K)$ be the least size of a finite catalogue $\mathcal{U} \subset \{0, 1\}^\mathbb{N}$ such that

$$(2.1) \quad \forall x \in K \ \exists u \in \mathcal{U} \ \forall k \in \{0, \dots, n\} : \quad F_u^k x \in Q^\delta.$$

The input assignment may depend on the actual state. A trajectory may leave and reenter Q , subject to every constraint in (2.1). Only the first n letters matter; infinite inputs make tail constructions unambiguous. Exact controlled feasibility will prove that this minimum is finite.

A precision sequence is positive, tends to zero, and satisfies $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. All logarithms are natural. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta > 1,$$

and put $a_* = \min_i a_i$, $a^+ = \max_i a_i$, $d = -\log a^+$, and $h = -\sum_i a_i \log a_i$. For later counting, set

$$H_b(v) = -v \log v - (1-v) \log(1-v), \quad \chi_a(v) = -v \log a_0 - (1-v) \log a_1,$$

with $0 \log 0 = 0$. Thus $H_b(a_0) = \chi_a(a_0) = h$.

2.1. Independent branch and radial perturbations. Define the sufficient threshold

$$(2.3) \quad \nu_0 = \min \left\{ \frac{a_*}{1000}, \frac{(\beta-1)d}{4(1+2/a_*)} \right\} > 0.$$

Fix $0 < \nu < \nu_0$ and three independent functions $\eta, \xi_0, \xi_1 \in C_c^\infty(B(0, 2))$, each zero at the origin, such that

$$(2.4) \quad \max_{f \in \{\eta, \xi_0, \xi_1\}} \max_{|\alpha| \leq 2} \|\partial^\alpha f\|_\infty \leq \nu.$$

Here $B(0, 2)$ is the fixed Euclidean open ball in $\mathbb{R}^2$. At each input state (s, z) put

$$(2.5) \quad p_0 = a_0 + \eta, \quad p_1 = a_1 - \eta, \quad R_i = a_i^\beta e^{\xi_i},$$

and define on the entire plane

$$(2.6) \quad \begin{aligned} F_0(s, z) &= \left(sR_0, \frac{R_0}{p_0}(z + p_1 s) \right), \\ F_1(s, z) &= \left(sR_1, \frac{R_1}{p_1}(z - p_0 s) \right). \end{aligned}$$

There is no imposed relation $R_i = p_i^\beta$. The matching at the origin concerns the matrices

$$(2.7) \quad A_0 = \begin{pmatrix} a_0^\beta & 0 \\ a_0^{\beta-1} a_1 & a_0^{\beta-1} \end{pmatrix}, \quad A_1 = \begin{pmatrix} a_1^\beta & 0 \\ -a_1^{\beta-1} a_0 & a_1^{\beta-1} \end{pmatrix}.$$

Theorem 2.2 (Common-cost boundary under asymptotic matching). *For every triple satisfying (2.3)–(2.4), the maps (2.6) are global C^∞ diffeomorphisms, fix the origin with $DF_i(0) = A_i$, drive bounded initial sets uniformly exponentially to the origin under all inputs, and make Q controlled invariant. For every fixed positive-area compact $K \subset Q$ and every precision sequence of finite exponent $\gamma \geq 0$,*

$$(2.8) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 \leq \gamma \leq \beta h$ the limit exists and equals γ/β .

For every $\zeta \in (0, 1)$ there is a fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma > \beta h$ and every precision sequence of that exponent,

$$(2.9) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = h < \liminf_n \frac{\log r_F(n, \delta_n; Q)}{n}.$$

This set may depend on the fixed system and ζ , but is chosen before the exponent, precision sequence, and horizon.

The threshold concerns shared rates across all positive-area compact sets. It does not classify the possible rates above the threshold or give an exact high-precision spectrum or limit for Q . The bound (2.3) is sufficient and is not claimed to be the largest allowable perturbation.

2.2. A different family: radius-autonomous mismatch. Keep a_0, a_1 as above, but now let $\beta_0, \beta_1 > 1$ and $\beta_{\max} = \max_i \beta_i$. For

$$(2.10) \quad 0 \leq \varepsilon < \frac{a_*}{4(\beta_{\max} + 1)}, \quad q_0(s) = a_0 + \frac{\varepsilon s}{1 + s^2}, \quad q_1(s) = a_1 - \frac{\varepsilon s}{1 + s^2},$$

define

$$(2.11) \quad \begin{aligned} G_0(s, z) &= (sq_0(s)^{\beta_0}, q_0(s)^{\beta_0-1}(z + q_1(s)s)), \\ G_1(s, z) &= (sq_1(s)^{\beta_1}, q_1(s)^{\beta_1-1}(z - q_0(s)s)). \end{aligned}$$

Set

$$(2.12) \quad \rho_i = a_i^{\beta_i}, \quad \chi = -\sum_i a_i \log \rho_i, \quad \gamma_0 = -\log \max_i \rho_i, \quad \rho_0^D + \rho_1^D = 1 \quad (D > 0).$$

Theorem 2.3 (Low-precision separation under radial mismatch). *Suppose $\beta_0 \neq \beta_1$. The maps (2.11) are global smooth diffeomorphisms with a common attracting origin under all inputs, and Q is controlled invariant. For every $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$ of area greater than $1 - \zeta$ such that, simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of exponent γ ,*

$$(2.13) \quad \lim_n \frac{\log r_G(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_G(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi, \quad D > h / \chi.$$

Every fixed positive-area compact $K \subset Q$ also has lower rate at least $\gamma h / \chi$ in this range.

Thus a fixed system can exhibit strict initial-set dependence at every sufficiently small positive exponent. The theorem is restricted to (2.11); it makes no claim about unequal powers in the independent-perturbation family. The precise matching criterion in this radius-autonomous class is given in Corollary 6.4.

3. COMPLETE FEASIBLE GRAPHS AND INDEPENDENT DISTORTION

Throughout this section the system is (2.6). Set

$$p_- = a_* - \nu, \quad p_+ = 1 - p_-, \quad r = e^\nu (a^+)^{\beta-1}, \quad t = \frac{e^\nu (a^+)^{\beta-1}}{1 - \nu/a_*}, \quad \sigma_0 = -1, \quad \sigma_1 = 1.$$

The following bounds apply on the whole plane: $p_- \leq p_i \leq p_+$, $R_i \leq r$, and $R_i/p_i \leq t$.

Lemma 3.1 (Global dynamics and one-step feasibility). *The dynamical assertions of Theorem 2.2 hold. For $s > 0$, write $y = z/s$, $P_i(s, y) = p_i(s, sy)$, and $W_i(s, y) = \xi_i(s, sy)$. There is a unique $\theta(s) \in (-1, 1)$ satisfying*

$$(3.1) \quad \theta(s) = 2P_0(s, \theta(s)) - 1.$$

The complete one-step feasible domains are $[-1, \theta(s)]$ for control 0 and $[\theta(s), 1]$ for control 1. Choosing 0 at the shared endpoint defines a natural exact code.

Proof. Since $\nu < a_*/2$, $-\log(1 - \nu/a_*) \leq 2\nu/a_*$. Hence

$$(3.2) \quad \log t \leq -(\beta - 1)d + (1 + 2/a_*)\nu < -\frac{3}{4}(\beta - 1)d < 0, \quad \log r < -\frac{3}{4}\beta d < 0.$$

Take

$$(3.3) \quad M = \frac{1+t}{1-t}, \quad \|x\|_* = \max(M|s|, |z|), \quad L = \max\{r, 2t/(1+t)\} < 1.$$

The second coordinate of (2.6) has coefficient at most t in absolute value on both z and s . Thus, on the entire plane,

$$(3.4) \quad \|F_i x\|_* \leq L \|x\|_*.$$

This bounds contraction toward the origin; it does not assert pairwise contraction. It also controls tails outside Q .

For global invertibility use the linear shear $J_i(s, z) = (s, z - \sigma_i s) = (s, v)$, whose norm and inverse norm are less than 2. Input and output shears of (2.6) give $(S, V) = (sR_i, vR_i/p_i)$, because $p_0 + p_1 = 1$. If $g_i = A_i^{-1}F_i - \text{id}$, then

$$J_i g_i(x) = (s\alpha_i(x), v\omega_i(x)), \quad \alpha_i = e^{\xi_i} - 1, \quad \omega_i = e^{\xi_i} a_i/p_i - 1.$$

This g_i vanishes outside $B(0, 2)$. On its support $|s| \leq 2$, $|v| \leq 2\sqrt{2}$, and differentiation gives

$$\begin{aligned} |\alpha_i| &\leq e^\nu \nu, & \|\nabla \alpha_i\|_2 &\leq e^\nu \sqrt{2} \nu, \\ |\omega_i| &\leq e^\nu \nu(1 + 1/p_-), & \|\nabla \omega_i\|_2 &\leq 2e^\nu \sqrt{2} \nu(1 + 1/p_-). \end{aligned}$$

For the last bound use $a_i/p_i \leq 2$. The two row-gradient estimates, followed by the inverse shear, yield

$$(3.5) \quad \|Dg_i\|_2 \leq 2e^\nu \nu [(1 + 2\sqrt{2}) + (8 + \sqrt{2})(1 + 1/p_-)] < 200\nu/a_* < 1/5.$$

The coarse bound follows from $e^\nu < 3$, $p_- > a_*/2$, and $a_* \leq 1/2$. It holds globally, since g_i has compact support. For each target $y \in \mathbb{R}^2$, the equation $F_i x = y$ is equivalent to $x = A_i^{-1}y - g_i(x)$, a contraction equation on the complete Euclidean plane. It has a unique solution. Also $DF_i = A_i(I + Dg_i)$ is nonsingular everywhere. The local inverse theorem now gives a smooth global inverse. Positivity of p_i makes (2.6) smooth, and the zero values of the perturbations give $DF_i(0) = A_i$; their first derivatives need not vanish.

In the calculation coordinates the actual map is

$$(3.6) \quad S = s a_i^\beta e^{W_i(s, y)}, \quad Y = \sigma_i + (y - \sigma_i)/P_i(s, y).$$

The physical set and metric are unchanged. On $0 < s \leq 1$, $|y| \leq 1$, the value at zero and the derivative bounds imply

$$(3.7) \quad \begin{aligned} |P_i - a_i|, |W_i| &\leq 2\nu s, \\ |(P_i)_s|, |(W_i)_s| &\leq 2\nu, \quad |(P_i)_y|, |(W_i)_y| \leq \nu s. \end{aligned}$$

The function $y - 2P_0(s, y) + 1$ has derivative at least $1 - 2\nu > 0$ and opposite signs at $-1, 1$, proving the unique divider. The conditions $-1 \leq Y \leq 1$ give exactly the asserted domains, each mapped angularly onto $[-1, 1]$. Since $S \leq rs < s$, there is no additional cap constraint. The vertex is fixed. These facts prove controlled invariance and the Borel exact code. $\square$

Lemma 3.2 (Moving graph cone and complete intervals). *Fix an actual initial radius $0 < s_0 \leq 1$. For every finite word w , the initial angles feasible at all times through $|w|$ form a nondegenerate closed interval $I_w(s_0)$. Its terminal angular map is increasing onto $[-1, 1]$, and its terminal radial graph $s = S(y)$ satisfies*

$$(3.8) \quad |S'(y)/S(y)| \leq 1.$$

Every complete prefix-free word family partitions the initial fibre, with disjoint interiors and all shared endpoints retained.

Proof. Suppose a parent graph obeys (3.8). Put $k = S'/S$ and

$$A_i = (P_i)_s S' + (P_i)_y, \quad B_i = (W_i)_s S' + (W_i)_y.$$

Equation (3.7) gives $|A_i|, |B_i| \leq 3\nu S$. Differentiating along this actual graph gives

$$(3.9) \quad q_i = 1 - (y - \sigma_i)A_i/P_i, \quad \frac{dY}{dy} = q_i/P_i, \quad k_{\text{new}} = P_i(k + B_i)/q_i.$$

The radial term B_i is independent of A_i . In particular $|q_i - 1| \leq 6\nu S/P_i$ and $q_i > 0$. Since $P_i(1 - P_i) \geq p_-(1 - p_-) \geq a_*/4$ and $(3P_i^2 + 6)\nu S \leq 9\nu < a_*/4$, we have

$$(3.10) \quad |k_{\text{new}}| \leq \frac{P_i(1 + 3\nu S)}{1 - 6\nu S/P_i} < 1.$$

On this graph $\Psi(y) = y - 2P_0(S(y), y) + 1$ has derivative at least $1 - 6\nu S > 0$ and opposite endpoint signs. It divides the parent once. Each feasible side is mapped increasingly onto the whole $[-1, 1]$, and (3.10) applies to its output graph.

The original fibre has $S \equiv s_0$ and $k = 0$. Induction proves the assertions at every time. Splitting each parent interval proves the partition assertion. The positive derivatives also give smooth dependence of finite-word endpoints on $s_0 > 0$ by the implicit function theorem. Closed endpoints and every intermediate constraint have been retained. $\square$

For a word $w = w_1 \cdots w_\ell$, set $\lambda_w = \prod_{j=1}^\ell a_{w_j}$, with empty product 1, and write

$$\mathcal{A}_w = \{(s_0, s_0 y) : 0 < s_0 \leq 1, y \in I_w(s_0)\} \cup \{0\}$$

for its complete physical feasible strip.

Lemma 3.3 (Separate radial and angular comparisons). *Fixed constants $C_R, C_A \geq 1$ satisfy, for every complete feasible word and every point of its initial interval,*

$$(3.11) \quad \begin{aligned} C_R^{-1} s_0 \lambda_w^\beta &\leq s_\ell \leq C_R s_0 \lambda_w^\beta, \\ C_A^{-1} \lambda_w^{-1} &\leq \partial_{y_0} Y_{w, s_0} \leq C_A \lambda_w^{-1}. \end{aligned}$$

Consequently

$$(3.12) \quad 2C_A^{-1} \lambda_w \leq |I_w(s_0)| \leq 2C_A \lambda_w,$$

and

$$(3.13) \quad C_A^{-1} \lambda_w \leq \text{area}(\mathcal{A}_w) \leq C_A \lambda_w.$$

Finite-word boundary curves have zero area and remain in the covers.

Proof. Along a complete feasible word $s_j \leq r^j s_0$. Set

$$\begin{aligned} B_P &= \frac{2\nu}{p_-(1-r)}, \quad B_R = \frac{2\nu}{1-r}, \quad d_0 = \frac{6\nu}{p_-} < 1, \\ B_q &= \frac{6\nu}{p_-(1-d_0)(1-r)}, \quad C_R = e^{B_R}, \quad C_A = e^{B_P+B_q}. \end{aligned}$$

Equations (3.7) and (3.9) imply

$$(3.14) \quad \begin{aligned} \left| \sum_{j<\ell} \log \frac{p_{w_{j+1}}(x_j)}{a_{w_{j+1}}} \right| &\leq B_P, \quad \left| \sum_{j<\ell} \xi_{w_{j+1}}(x_j) \right| \leq B_R, \\ \sum_{j<\ell} |\log q_{w_{j+1}}| &\leq B_q. \end{aligned}$$

For the last estimate use $|\log q| \leq |q - 1|/(1 - d_0)$. The identities

$$s_\ell = s_0 \lambda_w^\beta \exp \sum_{j<\ell} \xi_{w_{j+1}}(x_j), \quad \partial_{y_0} Y_{w, s_0} = \prod_{j<\ell} q_{w_{j+1}}/P_{w_{j+1}}$$

now prove (3.11). The two comparisons come from different error sums; there is no instantaneous power identity. The full terminal angular range in Lemma 3.2 gives (3.12). Integrating lengths against the physical Jacobian $s_0 ds_0$ gives (3.13), since $\int_0^1 s_0 ds_0 = 1/2$.

Each boundary has singleton fibre sections and smooth endpoints for $s_0 > 0$, hence zero area by Fubini. Their countable union and the vertex are null. $\square$

4. ARBITRARY APPROXIMATE INPUTS AND STOPPED WITNESSES

For any input u , put $V_F(n, \delta, u) = \{x \in Q : F_u^j x \in Q^\delta, 0 \leq j \leq n\}$. The exact natural code is a proof tool, not a restriction on this set. Define

$$\ell_- = \min_i \frac{a_i^\beta e^{-\nu}}{a_i + \nu} > 0, \quad C_b = \frac{2C_R C_A}{\ell_- (1 - 6\nu)}.$$

Lemma 4.1 (Physical first-deviation band). *Suppose an arbitrary input u covers $(s_0, s_0 y_0)$ through time n at tolerance δ and first differs from its natural code at time $j \leq n$. For $a = u|_{j-1}$ there is a separating point $b_a(s_0) \in I_a(s_0)$, depending only on the common prefix and radius, such that*

$$(4.1) \quad |y_0 - b_a(s_0)| \leq C_b \delta \lambda_a^{1-\beta} / s_0.$$

For fixed u, j these states lie on one side of that boundary, with physical area at most $C_b \delta \lambda_a^{1-\beta}$.

Proof. The common prefix is exactly feasible. On its full image graph let Y_* be the zero of $\Psi(Y) = Y - 2P_0(S(Y), Y) + 1$ and put $b_a = Y_{a, s_0}^{-1}(Y_*)$. For the wrong branch $i = u_j$, direct substitution in (3.6) gives

$$(4.2) \quad |Y_j| - 1 = |\Psi(Y)| / P_i, \quad |z_j| - s_j = S(R_i / P_i) |\Psi(Y)|.$$

At a tie both excesses are zero, so an alternate branch is included. A necessary condition for membership in Q^δ is $|z_j| - s_j < 2\delta$, by comparison with a point of Q at Euclidean distance less than δ . Use $S \geq C_R^{-1} s_0 \lambda_a^\beta$ from (3.11), the global lower bound $R_i / P_i \geq \ell_-$, and

$$\frac{d}{dy_0} \Psi(Y_{a, s_0}(y_0)) \geq (1 - 6\nu) C_A^{-1} \lambda_a^{-1}.$$

The mean value theorem proves (4.1). The wrong branch fixes its side. Multiplying the width by $s_0 ds_0$ and integrating cancels $1/s_0$, proving the area bound without a radial cutoff. Only the necessary condition at the first difference was used. Later reentry cannot remove it; no cone or exact-feasibility bound has been imposed on the subsequent approximate trajectory. $\square$

Let $\mathcal{L}(n, t)$ stop every binary branch at its first word with $\lambda_w \leq t$, or at depth n if there is no earlier crossing. Write $N(n, t) = \#\mathcal{L}(n, t)$ for $0 < t < 1$. This complete prefix-free counting tree is not assumed to be an optimal catalogue. Parent/child weight addition gives

$$(4.3) \quad \sum_{w \in \mathcal{L}(n, t)} \lambda_w = 1, \quad \lambda_w > a_* t, \quad N(n, t) \leq (a_* t)^{-1}.$$

Every proper prefix precedes a crossing, so the strict lower weight bound holds also for capped leaves.

Proposition 4.2 (Comparison with the actual catalogue). *For fixed constants B, C_0, C_U , all $n \geq 1$ and $0 < \delta < 1$,*

$$(4.4) \quad \frac{N(n, \delta^{1/\beta})}{1 + Bn} \leq r_F(n, \delta; Q) \leq N(n, (\delta/C_0)^{1/\beta}) \leq C_U \delta^{-1/\beta}.$$

Also $r_F(n, \delta; Q) \leq 2^n$.

Proof. For the lower bound set $t = \delta^{1/\beta}$ and take the physical witness $(1, y_w)$ at the midpoint of each $I_w(1)$ for the leaves at this threshold. Their intervals partition the fibre, and (3.12)–(4.3) give successive midpoint gaps greater than $2C_A^{-1}a_*t$. Each midpoint is interior to every ancestor, so has natural prefix w . An arbitrary input has at most one leaf as a prefix. Any other covered witness first differs at some $j \leq n$, with the one common prefix $a = u|_{j-1}$. It is a proper prefix of the witness leaf, so $\lambda_a > t$. Lemma 4.1, at radius 1, confines these witnesses to a one-sided band of width at most $C_b\delta t^{1-\beta} = C_bt$. Its spacing permits at most

$$B = \left\lceil 1 + \frac{C_A C_b}{2a_*} \right\rceil$$

witnesses for one input at that time. Adding the possible leaf-prefix witness and summing over times proves the lower bound.

For the upper bound set $t = (\delta/C_0)^{1/\beta}$ and use every leaf as an exact prefix, followed by the same infinite tail, say 0^∞ . Lemma 3.2 covers every radius and angle through the leaf, including all intermediate constraints and closed endpoints. A capped depth- n leaf needs no further constrained time. An earlier stop has radius at most $C_R t^\beta$, hence adapted norm at most $MC_R t^\beta$. With

$$C_0 = 2\sqrt{2}MC_R, \quad C_U = C_0^{1/\beta}/a_*,$$

the global bound (3.4) keeps the entire remaining tail within Euclidean norm $\delta/2$, strictly inside Q^δ at every later time. The vertex is fixed. Equation (4.3) gives the last inequality. Alternatively all exactly feasible length- n prefixes, extended arbitrarily, give the 2^n bound. $\square$

5. AREA-TYPICAL INITIAL STATES AND PROOF OF THE THRESHOLD

Let $c_k(x)$ denote the first k letters of the natural exact code, with the vertex assigned 0^∞ . The continuous maps and separating curves make every finite assigned-code set Borel measurable.

Lemma 5.1 (Physical typical sets). *For area-almost every $x \in Q$, the frequency of zero symbols in $c_k(x)$ tends to a_0 . Every compact $K \subset Q$ of positive area contains a compact positive-area set T on which this convergence is uniform. For every $0 < \vartheta < h$, there is $D_\vartheta \geq 1$ such that*

$$(5.1) \quad D_\vartheta^{-1}e^{-(h+\vartheta)k} \leq \lambda_{c_k(x)} \leq D_\vartheta e^{-(h-\vartheta)k} \quad (x \in T, k \geq 0).$$

For each $\zeta \in (0, 1)$ such a compact T can be chosen with $\text{area}(T) > 1 - \zeta$ inside Q .

Proof. Normalize area on Q , whose area is one. For every assigned word w , (3.13) bounds its probability by $C_A \lambda_w$. Thus the area of states whose length- k prefix has zero frequency at distance at least ϑ from a_0 is at most

$$(5.2) \quad C_A(k+1)e^{-2\vartheta^2 k}.$$

Indeed, for $t = j/k$, put $D_{\text{KL}}(t||a_0) = t \log(t/a_0) + (1-t) \log((1-t)/a_1)$. Then

$$\binom{k}{j} a_0^j a_1^{k-j} \leq e^{-k D_{\text{KL}}(t||a_0)}, \quad D_{\text{KL}}(t||a_0) \geq 2(t - a_0)^2.$$

The first inequality uses $\binom{k}{j} \leq e^{k H_b(j/k)}$: the binomial mass at j with parameter j/k is at most one. The endpoint types follow by continuity. The second inequality follows from $D_{\text{KL}}''(t) = 1/[t(1-t)] \geq 4$, with its value and first derivative zero at a_0 . Summing over types gives (5.2). Borel–Cantelli, followed by a countable sequence of positive ϑ tending to zero, proves almost everywhere convergence. No independence of the state-dependent digits has been used.

Remove the null set of finite-word boundary curves and the vertex. Egorov's theorem and inner regularity of area give a compact subset of any positive-area K with positive area and uniform frequency convergence. On Q they give a compact set of area greater than $1 - \zeta$. Since $-\log \lambda_{c_k(x)}/k$ is χ_a of the zero frequency, it converges uniformly to $\chi_a(a_0) = h$ on each chosen set. Absorb the finitely many early times into D_ϑ to obtain (5.1). All sets are chosen before any precision exponent or sequence. $\square$

Lemma 5.2 (Area covered by one arbitrary input). *If T satisfies (5.1), then for each $0 < \vartheta < h$ there is a constant C_ϑ such that, uniformly in u, n, δ and $0 \leq m \leq n$,*

$$(5.3) \quad \text{area}(T \cap V_F(n, \delta, u)) \leq C_\vartheta \left[e^{-(h-\vartheta)m} + \delta e^{(\beta-1)(h+\vartheta)m} \right].$$

Proof. Agreement through m confines the state to $\mathcal{A}_{u|m}$. If this agreement class meets T , its reference product is at most $D_\vartheta e^{-(h-\vartheta)m}$, and (3.13) gives the first term.

For a first disagreement at $j \leq m$, a nonempty class in T has common prefix $a = u|_{j-1} = c_{j-1}(x)$ for some $x \in T$. Equation (5.1) bounds its product from below by $D_\vartheta^{-1} e^{-(h+\vartheta)(j-1)}$. Lemma 4.1 bounds its area by $C_b D_\vartheta^{\beta-1} \delta e^{(\beta-1)(h+\vartheta)(j-1)}$. The sum over $j \leq m$ is a geometric sum with ratio greater than one, proving the second term. Empty classes give no contribution. Later disagreement times are already included in the agreement class through m . $\square$

Proof of Theorem 2.2. The dynamical assertions follow from Lemma 3.1. For a given positive-area compact K , choose $T \subset K$ as in Lemma 5.1. For small δ set

$$m = \min \left\{ n, \left\lfloor \frac{-\log \delta}{\beta(h+\vartheta)} \right\rfloor \right\}.$$

Then $\delta e^{\beta(h+\vartheta)m} \leq 1$, and Lemma 5.2 gives

$$\text{area}(T \cap V_F(n, \delta, u)) \leq 2C_\vartheta e^{-(h-\vartheta)m}.$$

Every catalogue for K must cover T , so

$$(5.4) \quad r_F(n, \delta; K) \geq \frac{\text{area}(T)}{2C_\vartheta} e^{(h-\vartheta)m}.$$

For $-\log \delta_n/n \rightarrow \gamma$, passage to rates and then $\vartheta \downarrow 0$ yields the lower bound $\min\{h, \gamma/\beta\}$. Proposition 4.2 and $K \subset Q$ give both upper bounds in (2.8). They coincide with the lower bound when $\gamma \leq \beta h$. The argument includes $\gamma = 0$ and equality at the threshold, and needs no monotonicity of δ_n .

For the high side choose once a compact T_ζ from Lemma 5.1 with area greater than $1 - \zeta$, and set $K_\zeta = T_\zeta \cup \{(0, 0)\}$. For every $\vartheta > 0$ and all sufficiently large n , all natural prefixes on T_ζ have zero frequency within ϑ of a_0 . The number of these prefixes is at most

$$(5.5) \quad (n+1) \exp \left[n \max_{\substack{0 \leq t \leq 1 \\ |t-a_0| \leq \vartheta}} H_b(t) \right].$$

Use each as an exact length- n prefix with any infinite extension, and one extra input for the vertex. This covers K_ζ exactly for all required times, for every $\delta > 0$. Continuity of H_b gives upper rate h . For every finite $\gamma > \beta h$, (2.8) gives lower rate h on this same set. Its rate therefore exists and equals h simultaneously for all those exponents and all their precision sequences.

It remains to obtain a strictly larger lower rate on Q . Since $a_0 \neq a_1$,

$$\Delta = \chi_a(1/2) - h = (a_0 - 1/2) \log(a_0/a_1) > 0.$$

For a given $\gamma > \beta h$, let

$$(5.6) \quad \theta_\gamma = \min \left\{ \frac{1}{2}, \frac{\gamma/\beta - h}{2\Delta} \right\}, \quad t_\gamma = (1 - \theta_\gamma)a_0 + \theta_\gamma/2.$$

By affinity of χ_a and strict concavity of H_b , $\chi_a(t_\gamma) < \gamma/\beta$ and $H_b(t_\gamma) > h$. Choose integers j_n with $j_n/n \rightarrow t_\gamma$. For all sufficiently large n , every word of length n with exactly j_n zeros has

$$-\log \lambda_w = n\chi_a(j_n/n) < -\log \delta_n/\beta.$$

Every proper prefix has still smaller cost. All these words are therefore terminal leaves of $\mathcal{L}(n, \delta_n^{1/\beta})$. The elementary type bound $\binom{n}{j} \geq e^{nH_b(j/n)}/(n+1)$ and Proposition 4.2 imply

$$(5.7) \quad r_F(n, \delta_n; Q) \geq \frac{\exp[nH_b(j_n/n)]}{(n+1)(1+Bn)}.$$

For completeness, the type bound follows since j is a mode of the binomial law with parameter j/n , so its mass is at least $1/(n+1)$; endpoint types are immediate. Taking lower rates in (5.7) yields $H_b(t_\gamma) > h$, proving (2.9). The strict gap may tend to zero as $\gamma \downarrow \beta h$. $\square$

Corollary 5.3 (Matched linear systems). *For*

$$A_0(s, z) = (a_0^\beta s, a_0^{\beta-1}(z + a_1 s)), \quad A_1(s, z) = (a_1^\beta s, a_1^{\beta-1}(z - a_0 s)),$$

the conclusions of Theorem 2.2 hold with the same threshold βh .

Proof. Set $\eta = \xi_0 = \xi_1 = 0$ in Theorem 2.2. The independent radial and angular constants are then $C_R = C_A = 1$. $\square$

6. RADIUS-AUTONOMOUS MATCHING AND MISMATCH

We now use (2.11), with parameters (2.10). The quantities in this section are independent of the feedback constants. For words let $\lambda_w = \prod_j a_{w_j}$ and $R_w = \prod_j \rho_{w_j}$. Write $\beta_{\min} = \min_i \beta_i$ and put

$$q_- = a_* - \varepsilon/2, \quad q_+ = 1 - q_-, \quad r_R = q_+^{\beta_{\min}} < 1, \\ C_R^P = \exp\left(\frac{\varepsilon}{q_-(1-r_R)}\right), \quad C_R^s = (C_R^P)^{\beta_{\max}}.$$

Lemma 6.1 (Actual strips and radial products). *The maps (2.11) are global smooth diffeomorphisms and admit a common origin-contracting norm on the whole plane. At a fixed initial radius, each complete feasible word has a closed angular interval $I_w(s)$, and every complete prefix-free tree partitions that fibre with endpoints included. Along a word set*

$$\mathcal{Q}_w(s) = \prod_j q_{w_{j+1}}(s_j), \quad \mathcal{R}_w(s) = \prod_j q_{w_{j+1}}(s_j)^{\beta_{w_{j+1}}}.$$

Then, uniformly in word and radius $0 \leq s \leq 1$,

$$(6.1) \quad (C_R^P)^{-1} \lambda_w \leq \mathcal{Q}_w \leq C_R^P \lambda_w, \quad (C_R^s)^{-1} R_w \leq \mathcal{R}_w \leq C_R^s R_w,$$

$s|_w = s\mathcal{R}_w(s)$ and $|I_w(s)| = 2\mathcal{Q}_w(s)$. Physical strip area is between $(C_R^P)^{-1} \lambda_w$ and $C_R^P \lambda_w$.

Proof. For $f(s) = s/(1+s^2)$, $|f| \leq 1/2$ and $|sf'| \leq 1/2$. Thus $g_i(s) = sq_i(s)^{\beta_i}$ has derivative

$$g_i'(s) = q_i^{\beta_i-1}(q_i + \beta_i s q_i') > 0,$$

with the bracket at least $a_* - (\beta_{\max} + 1)\varepsilon/2 > 0$. Since $q_i(s) \rightarrow a_i$ at infinity, g_i is onto $\mathbb{R}$. Its smooth inverse recovers s , then the positive coefficient $q_i^{\beta_i-1}$ recovers z . This proves global invertibility. With $M_R \geq 1$, $M_R > r_R/(1 - q_+^{\beta_{\min}-1})$, the norm $\max(M_R|s|, |z|)$ contracts relative to the origin by at most $\max\{r_R, q_+^{\beta_{\min}-1} + r_R/M_R\} < 1$.

At radius s , the angular maps are $T_{0,s}(y) = (y + q_1)/q_0$ and $T_{1,s}(y) = (y - q_0)/q_1$. The full domains are $[-1, \xi(s)]$, $[\xi(s), 1]$, where $\xi = q_0 - q_1$, with inverse branches $B_{0,s}(y) =$

$q_0(s)y - q_1(s)$ and $B_{1,s}(y) = q_1(s)y + q_0(s)$. The radius after the first letter does not depend on the angle, so

$$(6.2) \quad I_{iv}(s) = B_{i,s}(I_v(g_i(s))).$$

This induction retains all intermediate constraints and gives the partition and length assertions. Controlled feasibility follows.

The radial path obeys $s_j \leq r_R^j s$. Since $|q_i(s_j) - a_i| \leq \varepsilon s_j$, summing logarithmic errors proves (6.1); the radial error is at most $\beta_{\max}$ times the angular one. Integration against $s ds$ gives the area bounds. Finite-word boundary curves have zero area, and the vertex is fixed. $\square$

Lemma 6.2 (Relative radial-stop witness comparison). *Let $\mathcal{L}_R(n, t)$ stop words at first $R_w \leq t$, or at depth n , and let $N_R(n, t)$ be its size. Fixed constants $B_R, C_{0,R}$ satisfy*

$$(6.3) \quad \frac{N_R(n, \delta)}{1 + B_R n} \leq r_G(n, \delta; Q) \leq N_R(n, \delta/C_{0,R})$$

for all horizons and sufficiently small positive tolerances. For a first difference at time j , common prefix a , an arbitrary covered physical state lies in a one-sided band of width

$$(6.4) \quad |y - b_a(s)| \leq C_{b,R} \delta \lambda_a / (s R_a), \quad C_{b,R} = 2C_R^P C_R^s \max_i (a_i / \rho_i).$$

Its physical area is at most $C_{b,R} \delta \lambda_a / R_a$.

Proof. The common word is exact. Its angle map is affine with slope $Q_a(s)^{-1}$. For the differing branch i , the excess at the first difference is

$$|y_j| - 1 = |y - b_a(s)| / Q_{ai}(s), \quad s_j = s \mathcal{R}_{ai}(s).$$

The necessary condition $|z_j| - s_j < 2\delta$ and (6.1) give (6.4). The side is fixed by the differing input; integration cancels $1/s$. This includes ties and subsequent reentry.

For the lower bound use midpoint witnesses at radius 1 for leaves stopped at $t = \delta$. Write $\rho_* = \min_i \rho_i$. Every stopped or capped leaf satisfies $R_w > \rho_* t$. For any descendant $w = av$ of a fixed proper prefix a , $\lambda_v \geq R_v = R_w / R_a > \rho_* t / R_a$, because $\beta_i > 1$. By Lemma 6.1, all leaf intervals below a have length greater than $2(C_R^P)^{-1} \lambda_a \rho_* t / R_a$. Their ordered midpoints have at least that spacing. For one input and one first-difference time, (6.4) at $s = 1$ confines these descendant witnesses to a band of width $C_{b,R} \delta \lambda_a / R_a$. Hence at most

$$B_R = \left\lceil 1 + \frac{C_{b,R} C_R^P}{2\rho_*} \right\rceil$$

can be covered at that time. Add the one possible leaf prefix of the input and sum over times to get $1 + B_R n$. This is a conditional spacing comparison within the shared prefix, rather than a global minimum-length assertion.

For the upper bound, the full leaf intervals of (6.2) cover each physical fibre exactly through the leaf. Append the same infinite tail to each leaf at $t = \delta / C_{0,R}$, with $C_{0,R} = 2\sqrt{2} M_R C_R^s$. An early terminal radius is at most $C_R^s t$, and the global origin-contraction norm of Lemma 6.1 keeps the entire tail within Euclidean distance $\delta/2$ of the origin. Capped leaves are exactly feasible through n . All endpoints and the vertex are covered, proving (6.3). $\square$

Proof of Theorem 2.3. Lemma 6.1 proves the dynamical assertions. Write $V_G(n, \delta, u) = \{x \in Q : G_u^j x \in Q^\delta, 0 \leq j \leq n\}$ for the actual all-time initial set. For $0 < \gamma < \gamma_0$, the bound $R_w \leq e^{-\gamma_0 n}$ at depth n implies that every branch crosses either of the thresholds in (6.3) by time n , eventually for any precision sequence of exponent γ . For an uncapped radial stopping tree, replacement of a parent weight R_w^D by its children preserves total weight, since $\rho_0^D + \rho_1^D = 1$. All leaves have $\rho_* t < R_w \leq t$, and thus

$$(6.5) \quad t^{-D} \leq N_R(t) \leq (\rho_* t)^{-D}.$$

The comparison proves the full-set limit γD .

By the area bound of Lemma 6.1, the type probabilities of actual assigned codes are bounded by C_R^P times their Bernoulli weights. The proof of Lemma 5.1 therefore gives one uniformly typical compact T_ζ of area greater than $1 - \zeta$. Set $K_\zeta = T_\zeta \cup \{0\}$ before choosing any exponent, sequence, or horizon. On this same set, for each $\vartheta > 0$ a constant $A_\vartheta \geq 1$ satisfies

$$(6.6) \quad \begin{aligned} A_\vartheta^{-1} e^{-(h+\vartheta)k} &\leq \lambda_{c_k(x)} \leq A_\vartheta e^{-(h-\vartheta)k}, \\ A_\vartheta^{-1} e^{-(\chi+\vartheta)k} &\leq R_{c_k(x)} \leq A_\vartheta e^{-(\chi-\vartheta)k}. \end{aligned}$$

Both follow from the same uniform frequency convergence. For any positive-area compact K , the same construction supplies a positive-area compact $T \subset K$ with these bounds.

Agreement with an arbitrary input through m has area at most $C_R^P A_\vartheta e^{-(h-\vartheta)m}$. A nonempty first-difference class at time $j \leq m$ has a common typical prefix. Equations (6.4) and (6.6) bound its area by a constant times $\delta e^{(\chi-h+2\vartheta)(j-1)}$. Since $\chi > h$, summation gives, uniformly in the input,

$$(6.7) \quad \text{area}(T \cap V_G(n, \delta, u)) \leq A'_\vartheta [e^{-(h-\vartheta)m} + \delta e^{(\chi-h+2\vartheta)m}].$$

For $m = \min\{n, \lfloor (-\log \delta)/(\chi + \vartheta) \rfloor\}$, the second term is at most $e^{-(h-\vartheta)m}$. Since $\gamma < \gamma_0 \leq \chi$, covering the positive area of T yields lower rate $\gamma h/\chi$, after $\vartheta \downarrow 0$.

For the matching upper rate on K_ζ , use all radial stopping prefixes realized on T_ζ , with the full tails from Lemma 6.2. Put $T = -\log \delta$ and $c_{\max} = -\log \rho_*$. Each such stop has length at most

$$\ell \leq \frac{T + \log C_{0,R} + c_{\max} + \log A_\vartheta}{\chi - \vartheta}.$$

At a fixed length ℓ , distinct realized words have $\lambda_w \geq A_\vartheta^{-1} e^{-(h+\vartheta)\ell}$, whereas all length- ℓ word weights sum to one. Their number is at most $A_\vartheta e^{(h+\vartheta)\ell}$. Summing up to the displayed length, and adding one input for the apex, gives upper rate $\gamma(h+\vartheta)/(\chi-\vartheta)$. Taking $\vartheta \downarrow 0$ proves the second limit in (2.13), on one fixed K_ζ for all the stated sequences.

Finally let $b_i = \rho_i^D$, so $b_0 + b_1 = 1$. Then

$$(6.8) \quad D\chi - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

The inequality follows from $-\log t \geq 1 - t$; equality holds only when $a_i = b_i$ for both digits. This would imply $\beta_0 D = \beta_1 D = 1$, which is impossible under mismatch. Hence $D > h/\chi$. The root and the strict entropy comparison are classical counting, used only after the physical catalogue comparison has been established. $\square$

Proposition 6.3 (Matched radius-autonomous subfamily). *If $\beta_0 = \beta_1 = \beta$, the system (2.11), on its full range (2.10), has all the rate conclusions of Theorem 2.2, including the single compact set for every finite high-side exponent.*

Proof. Here $R_w = \lambda_w^\beta$ and $\mathcal{R}_w = \mathcal{Q}_w^\beta$. Lemma 6.1 gives the full intervals and area bounds required in Lemma 5.1. Equation (6.4) becomes a one-sided physical band of area at most $C_{b,R} \delta \lambda_a^{1-\beta}$. Equation (6.3) becomes (4.4) with fixed constants and angular stopping thresholds $\delta^{1/\beta}$ and $(\delta/C_{0,R})^{1/\beta}$. Its upper count follows from (4.3); the 2^n exact bound also holds. Thus the proofs of Lemma 5.2 and Theorem 2.2 use exactly these already derived estimates: choose the same m for the lower bound, one uniformly typical compact before all exponents for the high-side upper bound, and the type t_γ in (5.6) for the full-set strict lower bound. Every inequality and tail needed for those deductions has been supplied in this section. $\square$

Corollary 6.4 (Matching criterion within the radius-autonomous class). *For a fixed system (2.11), there is a nonempty interval of exponents adjacent to zero on which all*

fixed positive-area compact initial sets have the same rate for every corresponding precision sequence if and only if $\beta_0 = \beta_1$.

Proof. Under equality, Proposition 6.3 gives the interval $0 < \gamma \leq \beta h$. Under inequality, Theorem 2.3 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. $\square$

This criterion concerns only (2.11). It does not give a necessity statement for (2.6), overlapping feasible branches, or general smooth perturbations.

7. INDEPENDENT FACTORS BEYOND COORDINATE REDUCTION

Some perturbations, including the zero triple, reduce to old forms. The next construction excludes this explanation for genuine instances of (2.6). For comparison define the exactly matched rational feedback family

$$(7.1) \quad 0 \leq \varepsilon < \frac{a_*}{16(\beta + 1)}, \quad \hat{p}_0(s, z) = a_0 + \frac{\varepsilon z}{1 + s^2 + z^2}, \quad \hat{p}_1(s, z) = a_1 - \frac{\varepsilon z}{1 + s^2 + z^2},$$

$$(7.2) \quad H_0 = (s\hat{p}_0^\beta, \hat{p}_0^{\beta-1}(z + \hat{p}_1 s)), \quad H_1 = (s\hat{p}_1^\beta, \hat{p}_1^{\beta-1}(z - \hat{p}_0 s)).$$

This family is not asserted to satisfy the compact-support hypothesis of Theorem 2.2. Nor is every system in (2.6) asserted to reduce to (7.2).

Proposition 7.1 (A feasible-image obstruction). *There are systems in the full class of Theorem 2.2 with $R_i \neq p_i^\beta$ which admit no simultaneous ambient homeomorphism preserving Q and conjugating both controls to any system (7.2) or (2.11), including the linear endpoints and allowing label exchange. The identities are required on Q , on a domain containing Q and its needed one-step images. In particular no C^1 or bi-Lipschitz coordinate change of this kind works for these instances.*

Proof. Let $E(v) = e^{-1/v}$ for $v > 0$ and $E(v) = 0$ otherwise, and put

$$\kappa(x) = \frac{E(49/16 - |x|^2)}{E(49/16 - |x|^2) + E(|x|^2 - 9/4)}.$$

Its denominator is positive. The function is one on the closed ball of radius $3/2$ and zero outside radius $7/4$, hence is smooth and compactly supported in $B(0, 2)$. Define

$$f(s, z) = \kappa(s, z) \sin(4\pi z), \quad C_f = \max\{1, \max_{|\alpha| \leq 2} \|\partial^\alpha f\|_\infty\},$$

$$\bar{\nu} = 1/4000, \quad u = \bar{\nu}/(2C_f), \quad d_1 = u/64, \quad \beta = 2, \quad a_0 = (1 + d_1)/2, \quad a_1 = (1 - d_1)/2.$$

Take $\eta = 0$, $\xi_0 = uf$, and $\xi_1 = -uf$. They vanish at zero and satisfy (2.4) with bound at most $\bar{\nu}/2$. Here $a_* > 0.49$ and $a^+ < 0.51$. The first term in (2.3) exceeds $\bar{\nu}$; so does the second, since $-\log 0.51 > 2/5$ and $4(1 + 2/0.49) < 24$. Thus this construction belongs to the theorem with $\nu = \bar{\nu}$.

Let $D_i = Q \cap F_i^{-1}Q$ and $J_i = F_i(D_i)$. The factors $p_i = a_i$ are constant, so the source angles at a terminal angle Y are $y_0(Y) = a_0 Y - a_1$ and $y_1(Y) = a_1 Y + a_0$. Every source in Q lies where $\kappa = 1$. For fixed Y , its output radius is

$$a_i^2 s \exp[(-1)^i u \sin(4\pi s y_i(Y))].$$

It increases strictly with $s \in [0, 1]$, because the derivative bracket is at least $1 - 4\pi u > 0$. Thus J_i is exactly the radial region $0 \leq S \leq U_i(Y)$, $-1 \leq Y \leq 1$, where

$$U_0(Y) = a_0^2 e^{u \sin(4\pi y_0(Y))}, \quad U_1(Y) = a_1^2 e^{-u \sin(4\pi y_1(Y))}.$$

The parts of their boundaries inside Q are precisely the upper radial arcs, for $-1 < Y < 1$; both lie below the cap $s = 1$. Their log-radius difference is

$$(7.3) \quad \mathcal{D}(Y) = 2 \log \frac{1 + d_1}{1 - d_1} + 2u \sin(2\pi(Y + d_1)) \cos(2\pi d_1 Y).$$

Its constant term is at most $8d_1 = u/8$. At $Y = -3/4, -1/4, 1/4, 3/4$ the values of $\sin(2\pi Y)$ have alternating signs $+, -, +, -$ and magnitude one. The error in replacing the sine/cosine factor in (7.3) by this sine is at most $2\pi d_1 + 2\pi^2 d_1^2 < 1/4$. Therefore $\mathcal{D}$ has the same alternating signs and at least three distinct interior zeros. It is analytic on a neighbourhood of $[-1, 1]$ and is not identically zero, so it has only finitely many zeros there. Consequently

$$(7.4) \quad \partial J_0 \cap \partial J_1 \cap \text{int } Q$$

is a finite set with at least three points.

For a system (2.11), the radial maps are increasing and independent of the angle, by Lemma 6.1. Its feasible images are triangles with cap radii $g_i(1)$. Their interior boundary arcs are either disjoint or coincide in an entire arc. Neither gives the intersection in (7.4).

For (7.2), put $\hat{P}_i(s, y) = \hat{p}_i(s, sy)$ and $\hat{p}_- = a_* - \varepsilon/2$. The one-step angle maps are increasing on each feasible domain. Indeed $|\partial_y \hat{P}_i| \leq \varepsilon s$ gives $\partial_y Y \geq (1 - 2\varepsilon/\hat{p}_-)/\hat{P}_i > 0$. The determinant in physical coordinates is

$$\hat{p}_i^{2\beta-2}[\hat{p}_i + \beta s(\hat{p}_i)_s + ((\beta - 1)z + \sigma_i s)(\hat{p}_i)_z] > 0.$$

To check positivity, for $f = z/(1 + s^2 + z^2)$ use $|sf_s| \leq 1$, $|zf_z| \leq 1/2$, and $|sf_z| \leq 1/2$; the derivative contribution is at most $3\beta\varepsilon/2 < \hat{p}_-$. At fixed output angle, the output radius therefore increases with source radius: in (s, y) coordinates its derivative is the positive Jacobian divided by $\partial_y Y$. For every s the feasible angular maps have full range $[-1, 1]$. The interior upper boundaries of the two images consequently come from the source cap $s = 1$.

On that cap $\hat{P}_0 = a_0 + \varepsilon y/(2 + y^2)$ is increasing in y , while $\hat{P}_1 = a_1 - \varepsilon y/(2 + y^2)$ is decreasing, since $(y/(2 + y^2))' = (2 - y^2)/(2 + y^2)^2 > 0$. The cap angle maps are increasing, and the cap radii are the same positive powers of these factors. For $\varepsilon > 0$, the two upper arcs are therefore respectively increasing and decreasing as functions of Y , and meet at most once. For $\varepsilon = 0$ their constant radii are distinct, since $a_0 \neq a_1$.

A simultaneous ambient homeomorphism taking Q onto Q maps each feasible domain and image to its target counterpart. It maps boundaries and the interior of Q bijectively, preserving the cardinality in (7.4); label exchange does not change it. The finite intersection with at least three points contradicts both target possibilities. Finally $p_i^\beta = a_i^2$, whereas $R_i = a_i^2 e^{\pm u f}$ is nonconstant on Q , verifying the failure of instantaneous matching. $\square$

The proposition is an existence statement for the displayed instances, not a nonconjugacy assertion for every triple in the main theorem. It excludes only simultaneous changes preserving Q and the actual control identities. It makes no claim about changes that alter the constraint, conjugate only one control, or target an unspecified larger model class.

8. LOCAL MATCHING DOES NOT SUFFICE

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It violates global injectivity by collapsing positive-area regions onto an invariant ray.

Proposition 8.1. *For every choice of a_0, a_1, β in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*

- (4) For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(8.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(8.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0 t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1 t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (8.1)–(8.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0 s, -\rho_0 s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 5.3 and (2.8) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.2, where all K_ζ are chosen inside a single fixed system. The collapse example

is a counterexample to sufficiency of local matching and uniform convergence; it is not a counterexample to the reversible theorem.

9. SCOPE AND CONSEQUENCES

The common-cost boundary concerns coverage of actual initial states. Below it, a positive-area typical portion gives the lower bound for every compact initial set, including sets with empty interior. Above it, rare terminal types force additional controls for Q , while one uniformly typical compact set excludes those deviations. The choice of that compact set precedes every precision exponent.

The main theorem removes the instantaneous relation between branch lengths and radial factors. Origin contraction makes the two error sums finite, but their comparison to the real catalogue still requires the complete feasible graphs and the physical first-deviation band. The $1+Bn$ witness bound is what prevents arbitrary approximate controls, including those that reenter Q , from saving an exponential factor over the stopping count.

The mismatch theorem gives a different comparison. Its two radial first derivatives have unequal powers of the limiting branch proportions. Relative-prefix spacing connects the classical root D to the actual full-set rate, and uniformly typical physical sets attain the smaller rate. Necessity of equal powers is proved only in that radius-autonomous class. Neither theorem provides a general necessary and sufficient condition for feedback systems.

The perturbation class retains fixed compact support, small derivative bounds, matched first jets, and complementary feasible branches without interior overlap. These are global conditions: local matching and convergence alone fail by Section 8. Neither theorem gives a general classification or the full-set high-precision spectrum and its limit.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

REFERENCES

- [1] Z. Chen and X. Zhong, Invariance complexity and equi-invariability for control systems, *J. Math. Anal. Appl.*, **539** (2024), 128533. doi:10.1016/j.jmaa.2024.128533.
- [2] H. Chen, Y. Huang and X. Zhong, Variational principles of invariance entropy dimension, *J. Differential Equations*, **453**, Part 1 (2026), 113819. doi:10.1016/j.jde.2025.113819.
- [3] F. Colonius, Minimal bit rates and entropy for exponential stabilization, *SIAM J. Control Optim.*, **50** (2012), 2988–3010. doi:10.1137/110829271.
- [4] F. Colonius and B. Hamzi, Entropy for practical stabilization, *SIAM J. Control Optim.*, **59** (2021), 2195–2222. doi:10.1137/20M1367775.
- [5] I. Csiszár, The method of types, *IEEE Trans. Inform. Theory*, **44** (1998), 2505–2523. doi:10.1109/18.720546.
- [6] J. E. Hutchinson, Fractals and self-similarity, *Indiana Univ. Math. J.*, **30** (1981), 713–747. doi:10.1512/iumj.1981.30.30055.
- [7] C. Kawan and A. Da Silva, Invariance entropy for a class of partially hyperbolic sets, *Math. Control Signals Syst.*, **30** (2018), Article 18. doi:10.1007/s00498-018-0224-2.
- [8] C. Kawan, *Invariance Entropy for Deterministic Control Systems: An Introduction*, Lecture Notes in Mathematics, 2089, Springer, 2013. doi:10.1007/978-3-319-01288-9.
- [9] X. Nie, T. Wang and Y. Huang, Measure-theoretic invariance entropy and variational principles for control systems, *J. Differential Equations*, **321** (2022), 318–348. doi:10.1016/j.jde.2022.03.013.
- [10] T. Wang, Y. Huang and H.-W. Sun, Measure-theoretic invariance entropy for control systems, *SIAM J. Control Optim.*, **57** (2019), 310–333. doi:10.1137/18M1197862.
- [11] R. Yang, E. Chen, J. Yang and X. Zhou, Bowen’s equations for invariance pressure of control systems, *SIAM J. Control Optim.*, **63** (2025), 1104–1128. doi:10.1137/23M1607684.